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- **Best of luck for the exam and happy learning!**

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Table of Contents

AWS Certified Machine Learning Engineer - Associate Course

MLA-C01

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What We'll Cover: Data Ingestion and Storage

- Types, formats, and properties of data
- Data warehouses, lakes, and lakehouses
- ETL Pipelines and Orchestration
- AWS Storage and Streaming Services

Amazon Kinesis

Amazon FSx

Amazon Simple Storage Service (Amazon S3)

Amazon Elastic Block Store (Amazon EBS)

Amazon Elastic File System (Amazon EFS)

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Data Transformation, Integrity, and Feature Engineering

- Elastic MapReduce (EMR)
- Handling missing, unbalanced, and outlier data
- Common data transformations
- SageMaker data processing and analysis features
- AWS Glue

Amazon EMR

Amazon SageMaker

AWS Glue

AWS Managed AI Services

Amazon Personalize

Amazon Polly

Amazon Rekognition

Amazon Q

Amazon Comprehend

Amazon Lookout for Equipment

Amazon Forecast

Amazon Lex

Amazon Kendra

Amazon Textract

Amazon Transcribe

Amazon Translate

Amazon Fraud Detector

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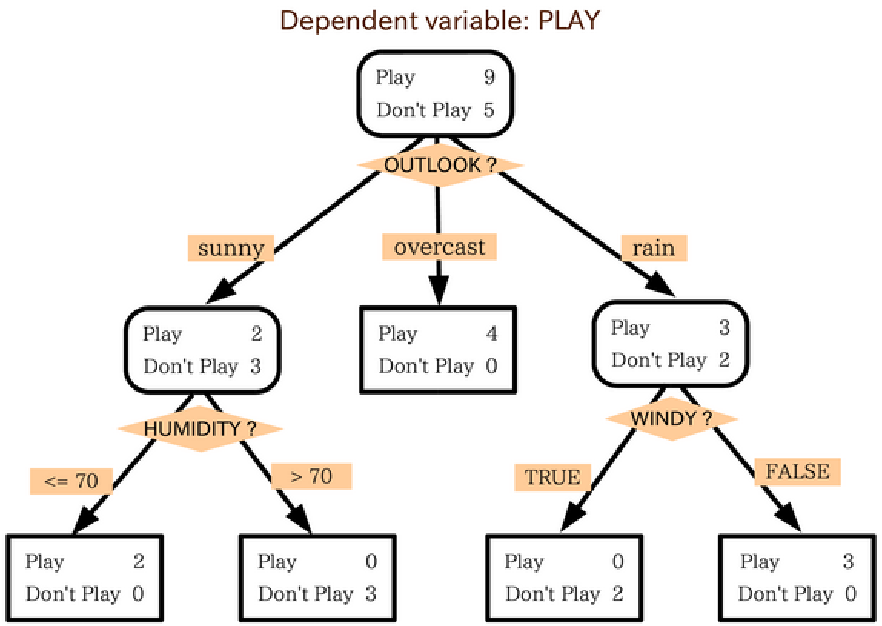
SageMaker Built-In Algorithms



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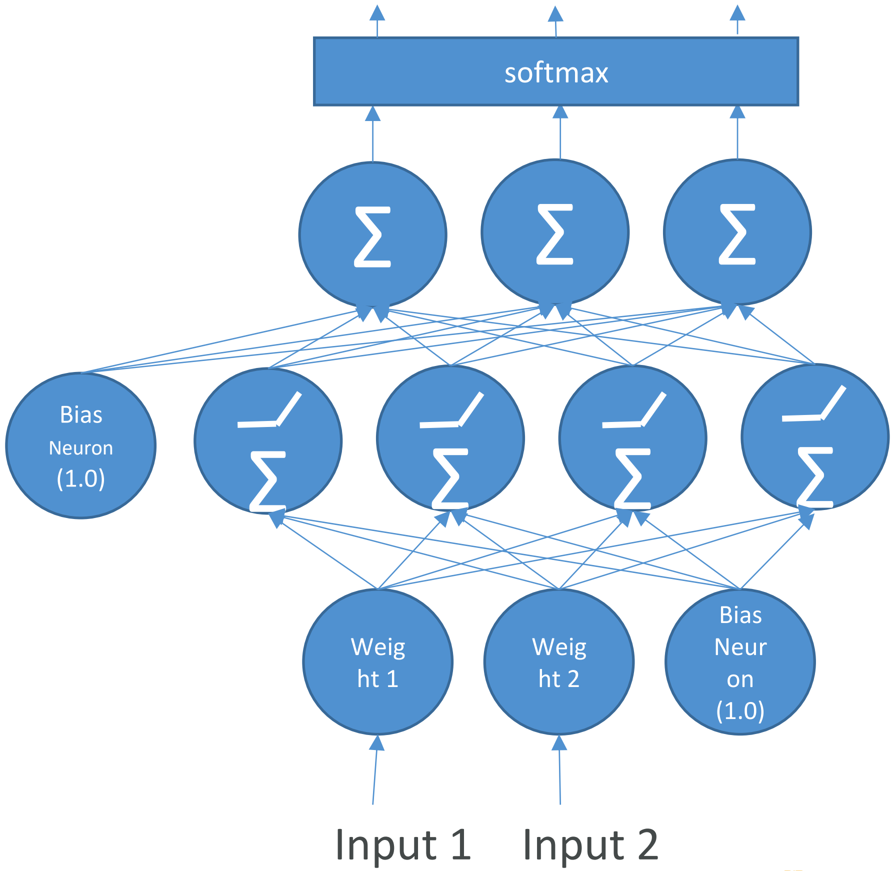
Model Training, Tuning, and Evaluation

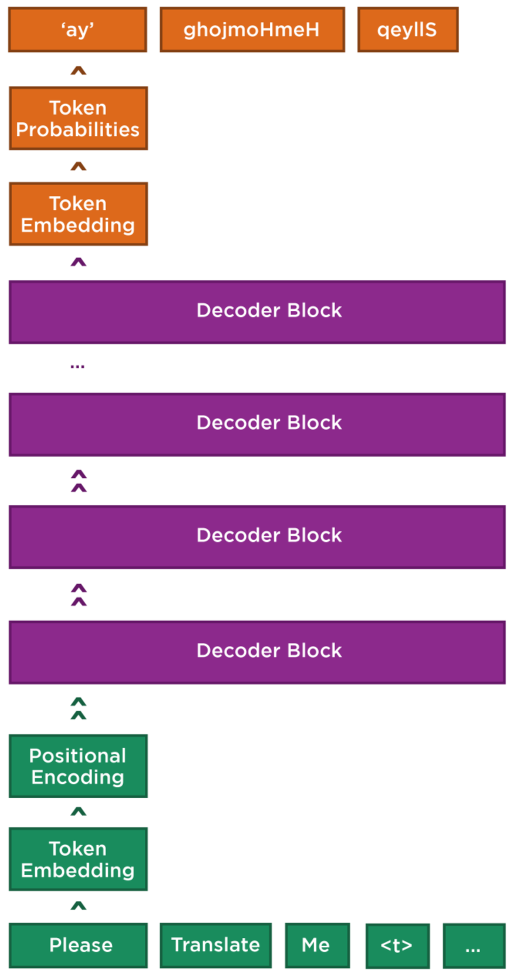
- Deep Learning Fundamentals
- Tuning techniques
- Measuring model performance
- Automatic Model Tuning with SageMaker

Amazon SageMaker



Generative AI Model Fundamentals

- The Transfomer Architecture
- Self-Attention
- How GPT Works
- SageMaker JumpStart
- Hands-on Labs



Building Generative AI Applications with Bedrock

- Foundation Models
- Retrieval-Augmented Generation (RAG)
- Knowledge Bases
- Vector Stores
- Guardrails
- LLM Agents
- Lots of hands-on labs

Amazon Bedrock

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Machine Learning Operations (MLOps)

- SageMaker in depth
- Amazon ECS
- Amazon ECR
- AWS CloudFormation
- AWS CDK
- AWS CodeDeploy
- AWS CodeBuild
- AWS CodePipeline
- Amazon EventBridge
- AWS Step Functions
- Amazon Managed Workflows for Apache Airflow (MWAA)

Amazon SageMaker

AWS Step Functions

Amazon EventBridge

Amazon Elastic Container Registry (Amazon ECR)

Amazon Elastic Container Service (Amazon ECS)

Amazon Managed Workflows for Apache Airflow (Amazon MWAA)

AWS CodeBuild

AWS Cloud Development Kit (AWS CDK)

AWS CodePipeline

AWS CodeDeploy

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Security, Identity, and Compliance

- Securing data in SageMaker
- AWS IAM
- KMS
- Macie
- Secrets Manager
- WAF
- Shield
- VPC
- PrivateLink

AWS Identity and Access Management (IAM)

AWS Key Management Service (AWS KMS)

AWS Secrets Manager

AWS Shield

Amazon Macie

AWS WAF

Amazon Virtual Private Cloud (Amazon VPC)

AWS PrivateLink

Management and Governance

Amazon CloudWatch

AWS CloudFormation

AWS Config

AWS CloudTrail

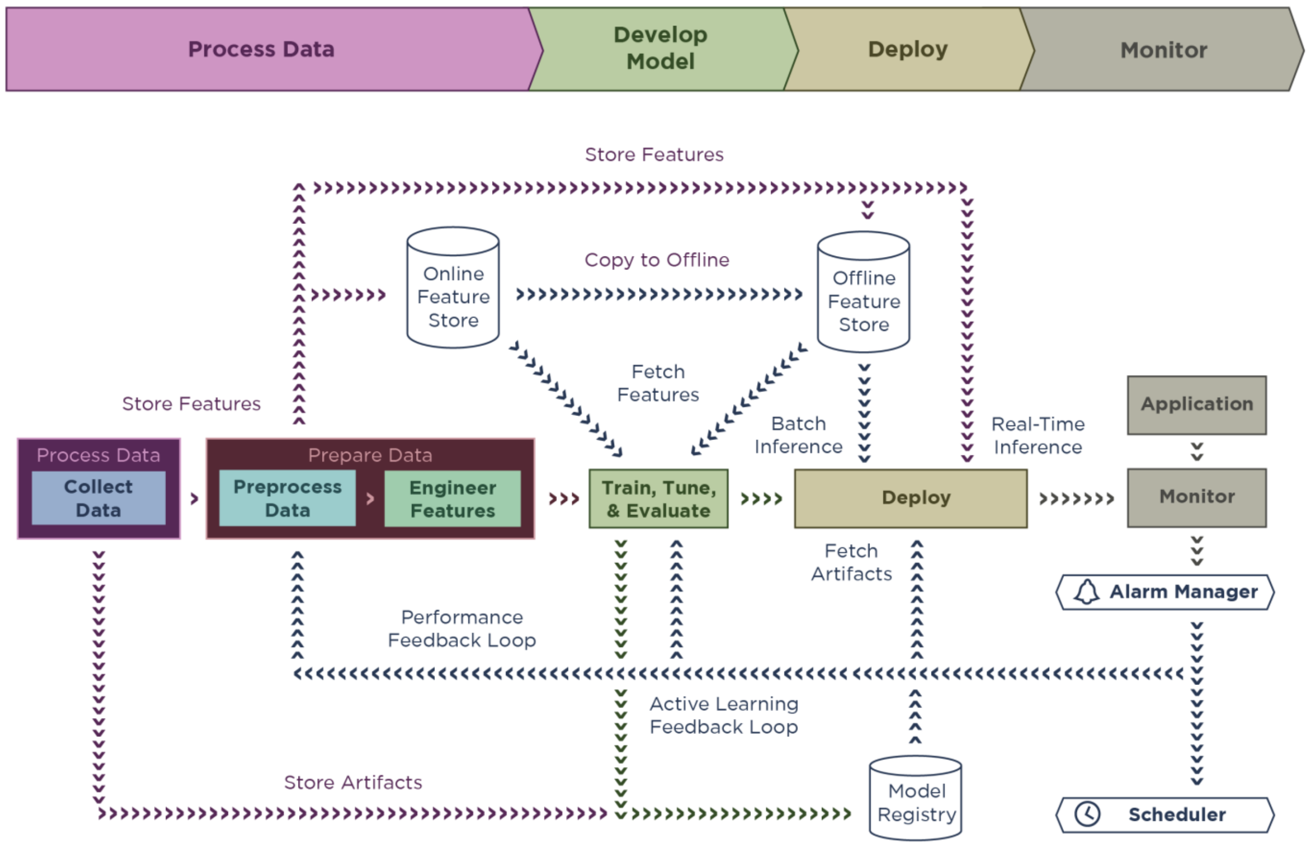
AWS X-Ray

AWS Trusted Advisor

AWS Budgets

AWS Cost Explorer

AWS Well-Architected Machine Learning Lens



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This all may seem familiar…

Target Candidate

- 1 Year experience with SageMaker and other ML Engineering services
- 1 year in software dev, devops, data engineering, or data scientist
- ML algorithms
- Data engineering fundamentals
- Software engineering and CI/CD
- NOT required:
  - Designing end-to-end ML solutions
  - Depth in NLP, Computer Vision,

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